0. **Code:** sirasanambeth2444 Sirasanambeti Himaja 2.zip

1. polynomial/polynomial/Program.cs

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```
namespace polynomial
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int[] coefficients = {
                5,
                6,
                7,
                8,
                9
            };
            int n = 2;
            int[] result = RaisePolynomial(coefficients, n);

            Console.WriteLine(string.Join(",", result));




        }
        public static int[] RaisePolynomial(int[] coefficients, int
          n)
        {
            if (n ≤ 0 || coefficients == null)
            {
                return null;
```

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1. **polynomial/polynomial/Program.cs**

```
        }
        //calculate the length of the return array
        int length = n * (coefficients.Length - 1) + 1;
        //copy all the values from the original array to end of the
          return array
        int[] array = new int[length];
        for (int i = 0; i < length; i++)
        {
            if (i < coefficients.Length)
            {
                array[length - coefficients.Length + i] = coefficients[i];
            }

        }
        for (int i = 0; i < n - 1; i++)
        {
            for (int j = 0; j < coefficients.Length; j++)
            {
                for (int k = 0; k < coefficients.Length; k++)
                {
                    if (j - k ≥ 0)
                    {
                        array[j] += array[j - k] * coefficients[k];
                    }
                }
        }
```

B3

int[] array2 = new int[array.Length];

reversed

array[j - (coefficients.Length - 1 - k)]

B4

B5

create new values.

array2[j]

needs to be shifted by the reversed. and you need to use a different array for calculations.

array = array2;

needs to be shifted by 2.

1. **polynomial/polynomial/Program.cs**

```
            }
            return array;
        }
    }
}
```